

Eprad eCNA Cinema Automation Plugin

Version 1.2 | Build 14

Use this plugin to monitor and control an Eprad eCNA Cinema Automation system from within your Q-SYS designs. The plugin communicates over two independent channels: the CAI (Cinema Automation Interface) ASCII TCP protocol for commands and status, and a direct HTTP connection for I/O flag detail and, optionally, Eprad QDC-400 dimmer lighting.

Upgrading from v1.1? Read this first: The Poll Rate and Scrape Rate properties have been renamed to Poll Interval and Scrape Interval, and now accept fractional seconds (e.g. 1.6) instead of whole seconds only. This was a deliberate rename, not an oversight — Q-SYS Designer does not reliably pick up a property *type* change (whole-number to fractional) on a component already placed in a design, but a new property name is treated as new and adopted automatically. The practical effect: after updating to v1.2, any existing instance of this plugin will show the two new properties at their defaults (30 and 2 seconds) rather than carrying over a custom value you'd previously set — you'll need to re-enter it once, but you should **not** need to remove and re-add the plugin itself. See [Poll Interval / Scrape Interval](#) below for details. Also see [What's New in 1.2](#) for the rest of this release.

- [What's New in 1.2](#)
- [Compatibility](#)
- [Configuration Overview](#)
- [Properties](#)
- [Controls](#)
- [Input & Output Flag Names](#)
- [QDC-400 Dimmer Lighting](#)
- [Control Pins](#)
- [Copyright](#)

What's New in 1.2

Version 1.2 adds optional monitoring and control of Eprad QDC-400 dimmer lighting boards, replicating the eCNA's own *Status: Light Control* web page, plus several changes to existing behavior:

- **QDC-400 Dimmer Lighting** — new optional Dimmer Control page with per-zone preset buttons (Up/Down/Mid1/Mid2/None), an offset fader, and animated per-channel level meters. See [QDC-400 Dimmer Lighting](#).
- **Poll Interval / Scrape Interval** — renamed from Poll Rate / Scrape Rate; now accept fractional seconds. See the callout above and [Properties](#).
- **Auto Refresh Names removed** — input/output names (and, when dimmer lighting is enabled, zone names and QDC board detection) are now fetched once on connect and otherwise only on demand. See [Refresh Setup Data](#).

- **Refresh buttons consolidated** — the separate Refresh Input Names / Refresh Output Names buttons (and, for dimmer lighting, two more) have been replaced by a single Refresh Setup Data button on the Setup page.

Compatibility

This plugin was developed and tested with Q-SYS Designer Software version 9.13.1 LTS, and is intended to be compatible with 9.4.8 LTS and newer. It has been tested against Eprad eCNA firmware v1.23 and v1.26, running on both the eCNA-5 and eCNA-10. It is designed to work with all eCNA models.

Note: The plugin communicates via the CAI TCP interface (default port 13002) and the eCNA HTTP web interface (port 80). Both must be reachable from the Q-SYS Core on your network. Because the eCNA supports only one CAI TCP connection per port at a time, avoid using the same port number in multiple concurrent designs.

Configuration Overview

Connect the Plugin

1. Drag the Eprad eCNA plugin into your Q-SYS schematic.
2. In the plugin Properties, configure the number of Macros and RDI devices for your installation. Enable Show Debug Tab only during commissioning. If the installation includes Eprad QDC-400 dimmer boards, enable Enable QDC Lighting and set the zone/channel counts — see [QDC-400 Dimmer Lighting](#).
3. Save the design to your Core and run (F5).
4. On the Setup page, enter the IP address of the eCNA and confirm the TCP Port (default 13002).

The plugin connects automatically once a valid IP address is entered. A Connection Status of **OK** confirms a successful connection.

Communication Architecture

The plugin uses two simultaneous communication channels:

- **CAI TCP (port 13002):** The primary control channel. Sends commands (EVT, OUT, RDI, MAC) and receives status responses (RST, RID) and unsolicited event reports (RPT,EVT). Polled at the RST Poll Rate interval as a safety net; most status updates arrive as unsolicited events. Only one device may have a port open at a time.
- **HTTP (port 80):** A secondary read-only channel for I/O flag detail, providing the state of all 44 output flags and 31 input flags, including house lights, curtain, masking, lens, sound, and aux outputs. Polled independently at the Web Scrape Rate.

Note: The HTTP scrape does not consume a CAI port. Multiple Q-SYS designs can scrape the eCNA web interface simultaneously without conflict. Only the CAI TCP connection is exclusive per port.

Unsolicited Event Reports

The plugin subscribes to unsolicited event reports from the eCNA (CFG, RPTON) on connection. The eCNA sends RPT, EVT messages when key state changes occur (START, STOP, CUE, projector power changes, fire stop, etc.). The plugin triggers an immediate RST status re-poll on receipt of these events, providing near-instant UI updates without relying solely on the poll timer.

Properties

Num Macros

Integer, 0–50 (default 50). Sets the number of Macro trigger buttons available on the Macros tab. Set to 0 to hide the Macros tab entirely.

Num RDI

Integer, 0–5 (default 5). Sets the number of RDI (Remote Device Interface) devices. Each device has 16 command buttons. Set to 0 to hide the RDI tab entirely.

Poll Interval

Decimal, 0.5–60 seconds (default 30). Interval between RST status poll commands sent over the CAI TCP connection. Because the plugin subscribes to unsolicited event reports, most real-time updates arrive without waiting for this timer. This rate primarily serves as a safety net to correct any state drift. Values of 30–60 seconds are appropriate for production use.

Note: Renamed from Poll Rate in v1.1, and changed from whole seconds to decimal seconds — see the upgrade notice near the top of this document.

Scrape Interval

Decimal, 0.5–60 seconds (default 2). Interval between HTTP requests to the eCNA web interface for I/O flag detail (and, when Enable QDC Lighting is on, dimmer status — see below). This is the primary update mechanism for the output and input flag indicators (house lights, curtain, masking, lens, sound, aux outputs, and input flags). Values below 2 seconds may interfere with direct operation of the eCNA device.

Note: Renamed from Scrape Rate in v1.1, and changed from whole seconds to decimal seconds — see the upgrade notice near the top of this document. The recommended/default value has also changed from 3 seconds to 2.

Show Debug Tab

Boolean (default off). When enabled, adds a Debug tab to the plugin showing Last TX, Last RX, and Last Error fields. Also enables verbose output to the Q-SYS debug window (visible via Help > Q-SYS Debug) showing all CAI commands sent, responses received, and status changes. Disable for production deployments.

Enable QDC Lighting

Boolean (default off). Turns on monitoring and control of Eprad QDC-400 dimmer lighting boards, adding the Dimmer Control page and its associated controls. See [QDC-400 Dimmer Lighting](#) for full details.

QDC Zones

Integer, 2–16 (default 2). Number of lighting zones to display on the Dimmer Control page — Zone 1 (House) and Zone 2 (Stage) are always present; zones above 2 are user-defined on the eCNA. This property is always shown regardless of the Enable QDC Lighting setting, but only takes effect once that property is turned on.

QDC Channels

Integer, 1–16 (default 2). Number of dimmer channel meters to display on the Dimmer Control page. A single QDC-400 board supports 4 channels; more than one board can be daisy-chained. This property is always shown regardless of the Enable QDC Lighting setting, but only takes effect once that property is turned on.

Controls

Setup Page

IP Address

Text entry. Enter the IP address of the eCNA. The plugin connects automatically when a valid address is entered or changed.

TCP Port

Text entry. The CAI TCP port number (default 13002). Change this to match the CAI channel assigned to this Q-SYS design. Common values: 13000 (channel 1), 13001 (channel 2), 13002 (channel 3), 13003 (channel 4).

Note: Port 13003 (channel 4) requires eCNA firmware version 1.25 or later.

Status LED / Status

Indicator LED and text field, read only. Displays the current connection state:

- **OK** (green): Connected and communicating.
- **Initializing** (blue): Connecting or establishing initial communication.
- **Disconnected / Fault** (red): Connection lost or error received.
- **Missing** (orange): No IP address configured.

Reconnect

Trigger button. Forces an immediate reconnection attempt. Useful if the connection has stalled, the IP address was changed, or the port was changed. Note: changing the TCP Port does not take effect until Reconnect is pressed.

Model

Text field, read only. Displays the eCNA device model returned by the RID command on connection.

Firmware

Text field, read only. Displays the eCNA firmware version returned by the RID command.

Refresh Setup Data

Trigger button. Re-fetches everything that's normally only fetched once, at connect: input flag names, output flag names, and — when Enable QDC Lighting is on — zone names, per-channel dimmer level/fade settings, and the QDC-400 board count. Use this after changing any of that data on the eCNA itself (Setup pages); day-to-day operation does not need it.

Note: New in v1.2, replacing the separate Refresh Input Names / Refresh Output Names buttons from v1.1 (and, for dimmer lighting, two more that briefly existed earlier in v1.2 development). This data changes rarely if ever after commissioning, so it is not automatically re-polled in the background.

Outputs Page

The Outputs page contains all eCNA output controls, grouped by type. Toggle buttons both send commands to the eCNA and reflect the current output state as returned by the web scrape.

Digital Projectors (Dig Proj1 Power, Dig Proj1 Video, Dig Proj2 Power, Dig Proj2 Video, Slide Projector)

Toggle buttons. Control and indicate the power and video output state of the two digital projectors and the slide/auxiliary projector. State is updated from the web scrape (flags 1–4, 19).

House Lights

Mutually exclusive Toggle buttons: None / Down / Mid1 / Mid2 / Up. Sends OUT HLxxx commands. State updated from scrape flags 5–8.

Stage Lights

Mutually exclusive Toggle buttons: None / Down / Up. Sends OUT SLxxx commands. State updated from scrape flags 9–10.

Curtain

Mutually exclusive Toggle buttons: None / Close / Open. Sends OUT CURTxxx commands. State updated from scrape flags 11–12.

Masking

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends OUT MSKxxx commands. State updated from scrape flags 13–15.

Lens

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends OUT LENxxx commands. State updated from scrape flags 16–18.

Sound

Mutually exclusive Toggle buttons: None / Aux1 / Aux2 / Dig1 / Dig2 / Mono / NonSync / SR / SVA. Sends OUT SNDxxx commands. State updated from scrape flags 20–28.

Mute

Independent Toggle button. Sends OUT MUTEON or OUT MUTEOFF. State updated from scrape flag 25.

Aux Output Flags 1–16

Toggle buttons arranged in two columns of eight. Each row shows the flag number (Out 1–Out 16), the toggle button, and the user-defined name fetched from the eCNA. Send `OUT OUTn0N/OFF` commands. State updated from scrape flags 29–44.

Note: Names refer to Output Flag names as configured in the eCNA web interface, not physical output names on any I/O board. To refresh names after changing them on the eCNA, use Refresh Setup Data on the Setup page.

Dimmer Control Page

Visible only when Enable QDC Lighting is on. Replicates the eCNA's own *Status: Light Control* web page, with animated level meters. See [QDC-400 Dimmer Lighting](#) below for full details on this page's controls and behavior.

Inputs Page

Displays all 31 eCNA input flags as LED indicators. Two columns: General Inputs (In1–In16) on the left and System Inputs on the right. Each General Input row shows the LED, the flag number (In1–In16), and the user-defined name fetched from the eCNA.

Note: Names refer to Input Flag names as configured in the eCNA web interface, not physical input names on any I/O board. Input flags are read-only indicators; they cannot be set from the plugin. To refresh names after changing them on the eCNA, use Refresh Setup Data on the Setup page.

Macros Page

Visible only when Num Macros is greater than 0. Contains trigger buttons for Macros 1 through N, arranged 10 per row. Each press sends `EVT MAC , n` to the eCNA. Text labels appear above each button.

RDI Page

Visible only when Num RDI is greater than 0. Contains trigger buttons for up to 5 RDI (Remote Device Interface) devices, each with 16 commands (labelled RDI 1-1 through RDI 5-16), arranged in two rows of 8 per device. Each press sends `EVT RDI , device , command` to the eCNA. Text labels appear above each button.

Program Page

Transport Controls (Start, Stop, Cue, Abort)

Trigger buttons. Send `EVT STA`, `EVT STP`, `EVT CUE`, and `EVT ABT` commands respectively.

Fault Controls (Set Fault, Clear Fault, Reset Fault, Cancel Alarm)

Trigger buttons. Send `EVT FLT`, `EVT CLR`, `EVT RES`, and `EVT CNL` commands respectively.

Program # / Set Prog / Set & Run

Combo box and trigger buttons. Select a program number (1–9) and press Set Prog to send `EVT PGM , n`, or Set & Run to send `EVT SPR , n` (set and immediately start).

Control State

Text field, read only. Displays the current control state from the RST response: **IDL** (idle) or **RUN** (running).

Stopped State

Text field, read only. Displays the stopped state: **OK**, **STP** (stopped), **FLT** (fault), or **FIR** (fire stop active).

Cue Number

Text field, read only. Displays the current cue number within the running program.

Program Number

Text field, read only. Displays the currently active program number.

Segment Name

Text field, read only. Displays the name of the current program segment.

Segment Index

Text field, read only. Displays the current segment index within the program.

CMD Access

Text field, read only. Displays the CAI command access status: **OK** (full access) or **BLK** (blocked — the eCNA is not accepting CAI commands).

Debug Page

Visible only when Show Debug Tab is enabled in Properties.

Last TX

Text field, read only. Displays the last CAI command string sent to the eCNA (without the trailing CR/LF).

Last RX

Text field, read only. Displays the last response line received from the eCNA.

Last Error

Text field, read only. Displays the most recent error message, including CAI errors (RSP,ERR), HTTP scrape errors, and internal runtime errors.

Input & Output Flag Names

Version 1.1 adds the ability to display user-defined names for the 16 General Input flags (In1–In16) and the 16 Aux Output flags (Out1–Out16). These names are configured on the eCNA web interface under Setup → Input Flag Names and Setup → Output Flag Names.

How Names Are Fetched

Input and output flag names are retrieved automatically from the eCNA on initial connection, and are not automatically re-fetched afterward — they very rarely change once a system is commissioned. If names

are changed on the eCNA later, use Refresh Setup Data on the Setup page to pick up the change.

Note: In v1.1, this was governed by an Auto Refresh Names property that periodically re-fetched names in the background. That property has been removed in v1.2 — see [What's New in 1.2](#).

Manual Refresh

The Refresh Setup Data button on the Setup page immediately refreshes input and output names (along with dimmer zone/channel data, if applicable), typically appearing within a few seconds.

Where Names Appear

- **Inputs page:** Each General Input row (In1–In16) shows the LED indicator, the flag number label, and the user-defined name in a text field alongside it.
- **Outputs page:** Each Aux Output Flag row shows the flag number, the toggle button, and the user-defined name in a text field to the right of the button.
- **Control pins:** In Names 1–16 and Out Names 1–16 expose the names as readable output pins for use elsewhere in the Q-SYS design.

Note: Names reflect the flag names as configured in the eCNA, not the physical label on any I/O board terminal. The eCNA supports up to 14 characters per name.

QDC-400 Dimmer Lighting

New in v1.2. When Enable QDC Lighting is turned on in Properties, the plugin adds a Dimmer Control page that monitors and controls Eprad QDC-400 dimmer boards, replicating the layout and behavior of the eCNA's own *Status: Light Control* web page.

Note: The CAI TCP protocol has no commands for Zone 3–16 lighting, individual dimmer channel levels, or the light level offset — only House and Stage lighting (already present on the Outputs page in v1.1) are reachable that way. All QDC-400 monitoring and control in this section is done through the eCNA's HTTP interface instead, sharing the same connection used for I/O flag detail.

Zones

Each configured zone (up to QDC Zones, set in Properties) is shown as a row with:

- **Zone name** — House and Stage (zones 1–2) are fixed; zones 3–16 are user-defined on the eCNA (Setup → Zone Names) and fetched automatically.
- **Preset buttons** — five mutually exclusive toggle buttons: Up, Down, Mid1, Mid2, None. Selecting one drives that zone's channels to the level configured for that preset on the eCNA (Setup → QDC-400 Dimmers), and resets the zone's offset to 0.
- **Offset fader and text box** — an adjustment from –100% to +100%, applied on top of the active preset's configured level for every channel assigned to that zone (clamped to 0–100%). When the active preset is None, the offset instead becomes the absolute level directly. Dragging the fader only sends a command to the eCNA once it has been still for about 1 second, to avoid flooding the connection while dragging; the displayed value updates immediately regardless.

Dimmer Channels

Each configured channel (up to QDC Channels, set in Properties) is shown as a vertical level meter with:

- **Level meter and percentage** — the channel's current output level, 0–100%.
- **Zone** — the name of the zone this channel is currently assigned to on the eCNA (Setup → QDC-400 Dimmers). More than one channel can be assigned to the same zone.
- **Present indicator** — a small LED that lights once the channel has reported a genuine non-zero level at least once. See [Dimmer Presence Detection](#) below.

Animated Level Meters

The eCNA can only be polled for dimmer status every couple of seconds at most (shared with the I/O flag scrape, which is intentionally kept fast and responsive since it is the most frequently used part of the plugin). A real dimmer fade often completes faster than that, so rather than have the meters visibly jump or lag behind, the plugin locally animates each channel toward its expected destination using the actual Level %/Fade-Time settings configured on the eCNA (Setup → QDC-400 Dimmers), the same values the physical dimmer itself uses.

This animation runs whenever a preset change is detected — whether it came from this plugin's own buttons, from the eCNA's own automation (a cinema server sending it a cue, a macro, another control system), or directly at the physical unit. House and Stage changes are typically detected within about a second via the same fast I/O flag scrape used for other outputs; other zones are detected on the next dimmer status poll.

The fade duration is the time to go from whichever preset was previously active to the newly selected one, regardless of how far apart their configured levels are — a short hop and a full 0–100% sweep at the same configured rate take about the same time. Offset adjustments always take about 3 seconds to reach their new level, regardless of which preset is active. Selecting None is effectively instant.

Note: This is a prediction, not a live readout, while a channel is animating — the plugin deliberately keeps showing its own smooth estimate rather than a possibly jumpy raw reading, and simply lets the animation finish on its own schedule. It resyncs with the eCNA's actual reported level as soon as the animation completes.

Dimmer Presence Detection

Not every QDC-400 channel necessarily has a physical dimmer connected and powered — for example during commissioning, or on a system with more channels wired than are actually driving fixtures. If a channel is expected to be at a non-zero level (based on its configured preset and offset) but the eCNA reports 0% two poll cycles in a row, the plugin assumes there's no working dimmer on that channel and stops trying to animate it — showing the honest 0% reading instead of repeatedly predicting a level that never arrives. A single genuine non-zero reading, at any time, immediately clears that assumption and lights the channel's Present indicator.

As a second, independent check, the plugin also scans the eCNA's Local I/O Network (Status → Local I/O Network) for active QDC-400 boards, shown as QDC Boards Found on the Dimmer Control page. If zero active boards are found, every channel is immediately marked as having no dimmer present, without

waiting on the per-channel detection above. This scan runs once on connect and otherwise only via Refresh Setup Data — it is not repolled in the background.

Control Pins

The following table lists the control pins available when this plugin is used in a Q-SYS schematic. All pins are optional; connect only those required by your design.

Pin Name	Value	String	Position	Direction
Setup				
IP Address		(text)		Input
TCP Port		(text)		Input
Connection Status	0 1 2 3 4 5	OK (Green) Compromised (Yellow) Fault (Red) Not Present (Grey) Missing (Orange) Initializing (Blue)	—	Output
Status LED	0–5	(same as Status)	—	Output
Connected	0 / 1	Off / On	0 / 1	Output
Reconnect	trigger	—	—	Input
Model		(text)		Output
Firmware Version		(text)		Output
Refresh Setup Data	trigger	—	—	Input
Program State				
Control State		IDL / RUN		Output
Stopped State		OK / STP / FLT / FIR		Output
Cue Number		(text)		Output
Program Number		(text)		Output
Segment Name		(text)		Output
Segment Index		(text)		Output
CAI CMD Access		OK / BLK		Output
Transport Commands				
Cmd Start	trigger	—	—	Input
Cmd Stop	trigger	—	—	Input
Cmd Cue	trigger	—	—	Input
Cmd Abort	trigger	—	—	Input
Cmd Fault	trigger	—	—	Input
Cmd Clear Fault	trigger	—	—	Input
Cmd Reset	trigger	—	—	Input
Cmd Cancel Alarm	trigger	—	—	Input
Set Program		(text / combo box, 1–9)		Input
Cmd Set Program	trigger	—	—	Input
Cmd Set And Run	trigger	—	—	Input
Macros (1 – Num Macros)				

Pin Name	Value	String	Position	Direction
Macros 1 ... Macros N	trigger	—	—	Input
RDI (1 – Num RDI × 16)				
RDI 1 ... RDI 80	trigger	—	—	Input
Outputs				
House Lights 1–5 (1=None 2=Down 3=Mid1 4=Mid2 5=Up)	0 / 1	Off / On	0 / 1	Input / Output
Stage Lights 1–3 (1=None 2=Down 3=Up)	0 / 1	Off / On	0 / 1	Input / Output
Curtain 1–3 (1=None 2=Close 3=Open)	0 / 1	Off / On	0 / 1	Input / Output
Masking 1–4 (1=None 2=Flat 3=Scope 4=Special)	0 / 1	Off / On	0 / 1	Input / Output
Lens 1–4 (1=None 2=Flat 3=Scope 4=Special)	0 / 1	Off / On	0 / 1	Input / Output
Sound 1–9 (1=None 2=Aux1 3=Aux2 4=Dig1 5=Dig2 6=Mono 7=NonSync 8=SR 9=SVA)	0 / 1	Off / On	0 / 1	Input / Output
Mute	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Power	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Video	0 / 1	Off / On	0 / 1	Input / Output
Dig2 Power	0 / 1	Off / On	0 / 1	Input / Output
Dig2 Video	0 / 1	Off / On	0 / 1	Input / Output
Aux Projector (Slide Projector)	0 / 1	Off / On	0 / 1	Input / Output
Aux Out 1–16	0 / 1	Off / On	0 / 1	Input / Output
Input & Output Flag Names				
In Names 1–16	(text — user-defined name for In1–In16)			Output
Out Names 1–16	(text — user-defined name for Out1–Out16)			Output
QDC-400 Dimmer Lighting (present only when Enable QDC Lighting is on)				
Zone Name 1 ... Zone Name (QDC Zones)	(text — zone name, House/Stage/user-defined)			Output
Zone Preset 1 ... Zone Preset (QDC Zones × 5) (5 per zone, in order Up/Down/Mid1/Mid2/None)	0 / 1	Off / On	0 / 1	Input / Output
Zone Offset 1 ... Zone Offset (QDC Zones)	-100 – 100	—	—	Input / Output
Zone Offset Text 1 ... Zone Offset Text (QDC Zones)	(text, editable, -100 to 100)			Input / Output
Channel Level 1 ... Channel Level (QDC Channels)	0 – 100	—	—	Output
Channel Level Text 1 ... Channel Level Text (QDC Channels)	(text, e.g. "75%")			Output
Channel Zone 1 ... Channel Zone (QDC Channels)	(text — zone name this channel is assigned to)			Output
Channel Present 1 ... Channel Present (QDC Channels)	0 / 1	Off / On	0 / 1	Output
QDC Board Count	(text — number of active QDC-400 boards found)			Output

Pin Name	Value	String	Position	Direction
Debug				
Last Tx		(text)		Output
Last Rx		(text)		Output
Last Error		(text)		Output

Copyright

MIT License

Copyright © 2026 Jay Wyatt and Steven Guttag

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the “Software”), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED “AS IS”, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.